



CONTACT

Italian, 28/09/1995

+39 3337196544 

github.com/mattiabestiaccia 

linkedin.com/in/mattia-bruscia

bruscia95@gmail.com 

INTERESTS

Artificial Intelligence &
Prompt Engineering UAVs
and Remote Sensing Human-
Robot Interaction Digital
Drawing

LANGUAGES

Native Italian
Professional English – B2

Willing to travel.
Driving licence available.

CONTEXT & PROMPT ENGINEER

MATTIA BRUSCIA

Master's student in Bioengineering at the University of Pisa, with a strong interdisciplinary background combining artificial intelligence, image/signal processing, and robotics. Experienced in MATLAB, Python, Arduino, and Java, applied in innovative projects from humanoid robotics to UAV image analysis. Recently expanded expertise to prompt engineering and Large Language Models (LLMs), with applications ranging from process optimization to AI-assisted coding (vibe coding) and research workflows. Skilled in the use of professional drone mission planning software (Pix4D, Agisoft Metashape) and AI platforms (OpenAI GPT, Anthropic models). Passionate about advancing the integration of AI into practical solutions across multiple domains.

SKILLS

- **Prompt Engineering & LLMs** — Expertise in designing, testing, and optimizing prompts for Large Language Models (LLMs) across diverse domains, including process automation, AI-assisted coding (vibe coding), and research workflows. Experienced with Anthropic, OpenAI, Hugging Face, n8n and LangChain/LangGraph- based architectures.
- **Programming & Frameworks** — Proficient in Python, MATLAB, Java, and Android development. Practical experience with TensorFlow, PyTorch, TensorBoard, and scikit-learn for deep learning, model evaluation, and experiment tracking. **Image & Signal Processing** — Skilled in UAV image analysis, feature extraction with Wavelet Scattering Transform, and deep learning for computer vision and classification tasks.
- **Human-Robot Interaction (HRI)** — Background in humanoid robotics, avatar-based interaction frameworks, and AI- driven control of facial expression systems.
- **Automation & Tools** — Experience in AI workflow automation using n8n and API-based integrations. Proficient with Pix4D, Agisoft Metashape, Fusion 360, and Tinkercad for UAV mission planning, 3D modeling, and simulation.

EDUCATION

MASTER'S DEGREE
IN BIOMEDICAL
ENGINEERING |
September 2021 –
Present

BACHELOR'S
DEGREE IN
BIOMEDICAL
ENGINEERING |
September 2014 –
2021

MORE EXPERIENCE

TUTOR

Tutoring university students
in engineering subjects.

LIFEGUARD

Beach lifeguard assistant,
later promoted to team
leader.

WAITER

Waiter for major events and
weddings.

EXPERIENCE

RESEARCH EXPERIENCES

RESEARCH FELLOW | UMCES, MARYLAND, USA | 2025

- Conducted research on UAV image analysis and vegetation **classification**.
- Developed multiscale feature extraction methods using **Wavelet Scattering Transform** (WST) for classification via Random Forests.
- Resulted in the forthcoming publication: "Multiscale Feature Extraction with Wavelet Scattering for UAV Vegetation Classification via Random Forests" (under review).

RESEARCHER & INTERN | E. PIAGGIO RESEARCH CENTRE, PISA | 2023 - 2024

- Worked on the humanoid robot Abel, focusing on **human-robot interaction** and integration of avatars and LLM-driven frameworks.
- Implemented deep learning algorithms (**CNNs**, inverse kinematics with regression models) for advanced facial robotics.
- Publications produced during this period:
 1. "A facial imitation framework for the simultaneous face control of a virtual avatar and a humanoid robot"
 2. "An Overview On Large Language Models Across Key Domains: A Systematic Review"
 3. "Advanced Control of Humanoid Facial Robotics: A Deep Learning Approach to Inverse Kinematics"

ENTERPRICE EXPERIENCES

AI EXPLORERS LAB | INTRODUCTION TO ARTIFICIAL INTELLIGENCE FOR KIDS | BOLZANO) | 2025

- Organized and conducted interactive workshops for elementary school children.
- Introduced concepts of **Artificial Intelligence**, **prompting**, and creative applications of **LLMs** in an engaging and accessible way.

AI CONTEXT ENGINEER | LEGODIGIT | TRENTO | 2025 - 2026

- **AI Developer Training**: Focus on the integration of AI tools, best practices for LLM implementation, and AI-assisted coding workflows.
- **Software Lifecycle Improvement**: Consultance on software maintenance, code refactoring, and the production of high-quality technical documentation.
- **Security & Auditing**: Conducted security audits and vulnerability assessments to safeguard system integrity.
- **Quality Assurance**: Developed and implemented robust automated test suites using Python to streamline debugging and enhance reliability.
- **Context Engineering**: Development of a scalable and actionable corporate knowledge base

CERTIFICAZIONI

LFS147: INTRODUCTION TO AI/ML TOOLKIT WITH KUBEFLOW

- Acquired foundational knowledge of AI/ML toolkits and workflows based on Kubeflow.
- Learned how to orchestrate and manage ML pipelines with Kubeflow on Kubernetes.
- Gained familiarity with best practices for scalable deployment of ML models and integration with open-source tools.
- Developed a solid understanding of Kubeflow's role in MLOps and cloud-native environments.